



# **TENUGHAT VIDYUT NIGAM LIMITED**

**तेनुघाट विद्युत निगम लिमिटेड**

**(A GOVT. OF JHARKHAND UNDERTAKING)**

JUPMI Building premises in ABD Area, Ranchi SmartCity,P.O&P.S.Dhurwa,Ranchi-04, (Jharkhand)

**NIT No.27/BMD/W/TVNL/RAN/2025-26**

**SRM RFx No-1000014214**



## ***TENDER DOCUMENT***

**SUBJECT: ANNUAL RATE CONTRACT for the two years and annual  
O/H of Boiler rotary equipment's (FD Fans, ID Fans, PA Fans &  
Seal Air Fans) of Unit-1&2.**



# Tenughat Vidyut Nigam Limited

तेनुघाट विद्युत निगम लिमिटेड

(A Govt. of Jharkhand Undertaking)

JUPMI Building premises in ABD Area, Ranchi SmartCity, P.O & P.S. Dhurwa, Ranchi-04, (Jharkhand)

Ref: 758/2026-27

Dated : 31.08.2026

## **NOTICE INVITING TENDER**

### **NIT No. 27/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1000014214**

On behalf of Tenughat Vidyut Nigam Limited (TVNL), Ranchi online tender is invited (in two parts Part-I - Technical & Commercial Bid, Part-II - Price Bid to be submitted in TVNL's website <https://webdispatcher.tvnl.in/srmprd> only) from experienced, reputed and financially sound firms for annual rate contract for 02 (Two) years and annual overhauling of Boiler rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) of Unit-1&2 of TTPS, Lalpania. Requisite cost of Tender Document & Earnest Money is to be submitted to designated account of TVNL well before submission deadline. Hard copy of Part-I-Technical & Commercial bid may be submitted in JUPMI Building premises in ABD Area, Ranchi Smart City, P.O & P.S. Dhurwa, Ranchi-04, (Jharkhand)\_TVNL Headquarter Ranchi on or before due date. The details of Tender are as follows: -

|                                                                                                        |                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description of Work                                                                                    | Annual Rate Contract for 02 (Two) years and annual overhauling of Boiler rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) of Unit-1&2 of TTPS, Lalpania. |
| Estimated Cost for two years (Rs.)                                                                     | Rs. 50,43,936.38 (Rupees Fifty Lakh Forty-Three Thousand Nine Hundred Thirty-Six and Thirty-Eight Paise) only plus GST extra as applicable.                          |
| Cost of Tender Documents (Non-Refundable) to be deposited through payment gateway link in TVNL website | Rs.10,000 /- (Rupees Ten thousand) only.<br>GST@18% Rs. 2,700/-<br>Total Tender fee Rs. /-12,700/-<br>(Rupees Twelve Thousand Seven Hundred) only.                   |
| Amount of EMD to be deposited in the form of BG/payment gateway link in TVNL website.                  | Rs.1,00,900/-<br>(Rupees One Lakh Nine Hundred) only.                                                                                                                |
| Bank Details for BG Creation                                                                           | Bank Name -ICICI Bank Ltd.<br>Name of Account -TENUGHAT VIDYUT NIGAM Ltd.<br>Branch Name-Lalpur, Ranchi<br>IFSC Code - ICIC0000292                                   |
| Downloading of Tender Paper                                                                            | From 02/09/2026                                                                                                                                                      |
| Last date & time of purchase tender paper                                                              | 23/09/2026 up to 14:00 Hrs.                                                                                                                                          |
| Last Date & Time for Submission of Tender                                                              | 24/09/2026 up to 14:00 Hrs.                                                                                                                                          |
| Date & Time of Opening of Part-I of Tender                                                             | 24/09/2026 at 16:00 Hrs.                                                                                                                                             |
| Place of submission of bid documents and address for communication                                     | ESE (C&M), TVNL, JUPMI Building premises in ABD Area, Ranchi Smart City, P.O.&P.S. Dhurwa, Ranchi-04, (Jharkhand)                                                    |

## **1. TRCHNICAL QUALIFICATION CRITERIA: -**

1.The Tenderer shall have to submit proof of experience of successfully executed work of ARC and Annual Overhauling of boiler Rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) or AMC of boiler and its auxiliaries in any Thermal Power Plant not less than 210 MW unit capacity under Govt./ Semi Govt. / CPSU/SPSU/ SEB's / Reputed Private Organization.

2.The Tenderer is required to submit copy (ies) of work orders along with successful executed work certificate of similar job in Thermal Power Station of unit capacity 210 MW or above during last five years as: -

i. Three similar completed works each costing not less than 40% of the estimated cost.

OR

ii. Two similar completed works each costing not less than 50% of the estimated cost.

OR

iii. One similar completed works costing not less than 80% of the estimated cost.

“Similar work means the bidder must have successfully executed working experience of ARC and Annual Overhauling of boiler Rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) or AMC of boiler and its auxiliaries in any Thermal Power Plant not less than 210 MW unit capacity under Govt./ Semi Govt. / CPSU/SPSU /SEB's/ Reputed Private Organization.

3.The firm has to submit a self-declaration undertaking that company has not been blacklisted from any organization like NTPC/BHEL/SEBs/any PSU at the time of participation in NIT.

4.The bidder should have valid GST Registration Certificate, Employee Provident Fund Code No. and Ownership/Partnership deed.

## **2. FINANCIAL QUALIFYING CRITERIA: -**

i. Average annual turnover of preceding 03 (three) financial years (i.e. 2023-24, 2024-25 & 2025-26) should not be less than 30% of the estimated cost. A certificate issued by Chartered Accountant/Auditor should be attached. Annual Accounts Reports of preceding 03 years certified by Auditor/CA for determining, overall profitability, average annual turnover etc, are to be submitted. In case where audited result for the last preceding financial year are not available, certificate of financial statements from a practicing Chartered Accountant shall also be considered acceptable.

ii. Solvency certificate should be current F.Y or F.Y 2025-26. The bidder should have financial credibility/ capacity of at least 10% of the estimated cost. Bidder shall submit Bank solvency certificate of any Indian Nationalised / Scheduled commercial Bank in support of their financial credibility/capacity.

iii. The latest copy of ITR return and GSTR 3B should be submitted.

3. The tender document is to be downloaded from the TVNL website [www.tvnl.in](http://www.tvnl.in).

Tenders will have to be submitted in TVNL Web Portal only website:

<https://webdispatcher.tvnl.in/srmprd> For easy reference “**Vendor Bidding Manual for Open Tender**” can be downloaded from the TVNL website homepage.

4. After advertisement of NIT, intended bidder will first communicate with TVNL contact person Kavindra Ram EEE(C&M) for this aforesaid NIT only on his Mobile No- No-983127953 and send their firm details well before the submission deadline. Also company's contact person name, email, Mobile no may be sent in email ID: [k.ram@tvnl.in](mailto:k.ram@tvnl.in), so that login ID & password can be created in SRM-SAP which will be sent to bidders contact person email. Bidder contact person has to register at TVNL website: <https://webdispatcher.tvnl.in/srmprd> with the given ID & password.

5. The cost of tender document and the Earnest Money shall be deposited well before the submission deadline and has to be deposited only through payment gateway link in TVNL website. (DD will not be acceptable). The copy of the transaction ID or receipt may be submitted to TVNL contact person in person or by mail so that the contact person will allow access to open the RFx details and initiate bidding at TVNL website and attach the detail of part – I.
6. Bidder have to fill as well as attach scanned copies of the related documents to qualify in the part-I (Technical & commercial) on TVNL website portal: <https://webdispatcher.tvnl.in/srmprd> and also the bidder may submit the hardcopy of bid part-I (Technical & commercial) to the ESE (C&M), TVNL, JUPMI Building Premises in ABD Area, Smart City, Dhurwa, Ranchi-834004, Jharkhand office along with the details of paid cost of tender document and paid or BG of Earnest money deposit (EMD).
7. Bidders have to bid for part-II (Price part) on TVNL website: <https://webdispatcher.tvnl.in/srmprd> only. **Price Part is not to be submitted in Hard copy** (in sealed envelope).
8. Bidder will not get access to bidding in SRM-SAP without providing the details of cost of tender document. Any type of tempering/alteration with tender document shall not be allowed and offer, if submitted so, will be summarily rejected.
9. Tender without Earnest Money shall not be considered. Cost of tender document and Earnest Money Deposit (EMD) of requisite value as specified above is to be deposited well before submission deadline. EMD may be furnished in the form of Bank Guarantee in required format from any Indian Nationalized Bank/ Schedule Commercial Bank in favour of “TENUGHAT VIDYUT NIGAM LIMITED” payable at Ranchi. The Bank Guarantee should remain valid for 12 months. In case BG is to be submitted against Earnest money deposit (EMD) it is to be submitted to TVNL contact person in a separate envelope so that the contact person will enter the BG details against EMD in the respective RFx well before the submission deadline, failing which offers are liable for rejection.
10. The Earnest Money will be forfeited on revocation of tender before expiry of validity of the tender or on refusal to enter into the contract after the award is made by TVNL to the bidder.
11. Any corrigendum/Addendum/Extension of bid opening date and cancellation of NIT will be published through our website [www.tvnl.in](http://www.tvnl.in) only and not through newspaper. The bidders are requested to visit our website regularly for any addendum/corrigendum/extension till opening of NIT.
12. The due date of bid opening as mentioned in the tender will remain firm. However, in case of unforeseen circumstances or due to administrative reasons the bids are not opened on the due date, the same shall be opened on the next working day at the same time. In case the last date of submission of bids is declared as holiday, the bids received up to the time of closing of bids on the next working day shall be considered for opening.
13. TVNL reserves the right to reject any / all offer increase/decrease or to distribute the quantity among the bidders without assigning any reason thereof.

(R. Ranjan)  
Electrical Superintending Engineer (C&M)



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### BILL OF QUANTITY

### PART- I (TECHNICAL & COMMERCIAL)

| Sl. No. | Particulars                                                                                                                                                                | Details to be furnished by the Bidder (Enclose separate sheets if required)                                                                                                                                                      | Flag No. It is mandatory | Page No. |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------|
| 1.      | Name of the Bidder (Enclose Letter Head of firm)                                                                                                                           |                                                                                                                                                                                                                                  | I                        |          |
| 2.      | Address of Bidder with Telephone, Fax, E-mail                                                                                                                              |                                                                                                                                                                                                                                  | II                       |          |
| 3.      | Name of the Proprietor/ Head of the Firm and its key personnel with organizational chart giving details of qualification and experience etc.                               |                                                                                                                                                                                                                                  | III                      |          |
| 4.      | Details of maintenance set up with resources like permanent employees (with designation & strength), tools & tackle and other equipment with capacities                    |                                                                                                                                                                                                                                  | IV                       |          |
| 5.      | Name and Qualification of the site-in-charge                                                                                                                               |                                                                                                                                                                                                                                  | V                        |          |
| 6.      | Work Experience of Bidder (Attach copy of work order of similar work and its successful completion certificate during last seven years as per S. N. 1. e. of QR)           | Work Order No.....<br>Dated.....<br>Completion Certificate<br>Dated.....<br>Work Order No.....<br>Dated.....<br>Completion Certificate<br>Dated.....<br>Work Order No.....<br>Dated.....<br>Completion Certificate<br>Dated..... | VI                       |          |
| 7.      | The latest copy of ITR return and GSTR 3B should be submitted.                                                                                                             |                                                                                                                                                                                                                                  | VII                      |          |
| 8.      | Financial Credibility/ Capacity (Attach latest Bank Solvency Certificate)                                                                                                  |                                                                                                                                                                                                                                  | VIII                     |          |
| 9.      | Average annual turnover of bidder during the last 03 financial years (Attach audited Balance Sheet & P/L Account for last three years ending 31 <sup>st</sup> March, 2026) | Average annual turnover of bidder during the last 03 (three) financial years Rs.....<br>Annual Turnover<br>FY 2023-24: Rs.....<br><br>FY 2024-25: Rs.....<br><br>FY 2025-26: Rs.....                                             | IX                       |          |

|     |                                                                                                      |                                                                    |        |  |
|-----|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------|--|
| 10. | Name & Designation of Authorized Signatory (Attach Power of Attorney if applicable)                  |                                                                    | X      |  |
| 11. | Nature of Firm (Proprietorship/Partnership /Pvt. Ltd./Public Sector) (Attach proof)                  |                                                                    | XI     |  |
| 12. | Permanent Account Number (PAN) (Attach proof)                                                        |                                                                    | XII    |  |
| 13. | GST Registration Number (Attach proof)                                                               |                                                                    | XIII   |  |
| 14. | EPF. Code No. (Attach copy of registration)                                                          |                                                                    | XIV    |  |
| 15. | Particulars of Earnest Money (Transaction ID No. /BG No. & Date)                                     |                                                                    | XV     |  |
| 16. | Particulars of Cost of Tender Document (BOQ) (Transaction Id. & Date)                                |                                                                    | XVI    |  |
| 17. | Whether the firm are a Govt. Organ. /Govt. U/T.                                                      |                                                                    | XVII   |  |
| 18. | Whether the firm is registered with NSIC/SSI (Jharkhand). (enclose copy of certificate)              |                                                                    | XVIII  |  |
| 19. | Specification of the material to be supplied:                                                        |                                                                    | XIX    |  |
| 20. | Quantity both in figure & in words:                                                                  |                                                                    | XX     |  |
| 21. | F.O.R (Prices Basis)                                                                                 |                                                                    | XXI    |  |
| 22. | Packing & Forwarding charges                                                                         |                                                                    | XXII   |  |
| 23. | Freight                                                                                              |                                                                    | XXIII  |  |
| 24. | GST (kindly indicate the taxes applicable)                                                           |                                                                    | XXIV   |  |
| 25. | Payment Terms for materials                                                                          |                                                                    | XXV    |  |
| 26. | Transit Insurance charges                                                                            | <b>The TVNL have its own marine policy with insurance company.</b> | XXVI   |  |
| 27. | Transportation charges                                                                               |                                                                    | XXVII  |  |
| 28. | Validity period of the offer (minimum 120 days )                                                     |                                                                    | XXVIII |  |
| 29. | Guarantee period of the material to be supplied                                                      |                                                                    | XXIX   |  |
| 30. | Whether you will deposit Security Money. If not reasons thereof with documentary proof of exemption: |                                                                    | XXX    |  |

|     |                                                                                                                                                                                                                                                                                                                                                        |  |        |  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|--|
| 31. | Whether you will furnish the Performance Guarantee                                                                                                                                                                                                                                                                                                     |  | XXXI   |  |
| 32. | Inspection                                                                                                                                                                                                                                                                                                                                             |  | XXXII  |  |
| 33. | Do you fulfill the pre-requisite qualification?                                                                                                                                                                                                                                                                                                        |  | XXXIII |  |
| 34. | Whether you accept all term & conditions of B.O.Q.                                                                                                                                                                                                                                                                                                     |  | XXXIV  |  |
| 35. | Penalty                                                                                                                                                                                                                                                                                                                                                |  | XXXV   |  |
| 36. | Details required for ERP Vendor Coding:-<br>a. Name of firm :<br>b. Address :<br>PIN:<br>c. E-mail ID :<br>d. Phone No. :<br>e. Name of Contact Person:<br>f. GST No. :<br>g. PAN No :<br>Details required for EMD Refund through RTGS/NEFT :<br>h. Bank Name :<br>i. Branch :<br>j. Branch Code :<br>k. A/c. No. :<br>l. IFS Code :<br>m. MICR Code : |  | XXXVI  |  |
| 37. | Other details and remarks, if any                                                                                                                                                                                                                                                                                                                      |  | XXXVII |  |

(Signature & Seal of Bidder)

**NOTE: - DETAILS SHEET/DOCUMENTS SUBMITTED ALONG WITH BID SHOULD BE FLAGGED. IT IS TREATED AS NECESSARY FOR SUBMISSION OF TENDER DOCUMENTS.**



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JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S.Dhurwa, Ranchi-04, (Jharkhand)

#### **INSTRUCTIONS TO BIDDERS FOR ONLINE BIDDING**

Each of these instructions must be read carefully and followed strictly. Failure in complying with any of these would render your offer liable for non-submission and TVNL or its any department will not take any responsibility for any bid not successfully Submitted/Participated in **RFx** (Request **for Quotation**).

#### **IMPORTANT STEPS FOR PARTICIPATION IN BIDDING:**

If Bidder is interested to participate in the Open tender, then follow the below steps to provide the bid quotation. Note: Use Only Internet Explorer browser to login into TVNL Portal.

- 1) Open RFx
- 2) Pay tender Fee & Earnest Money Deposit Amount and inform the same to TVNL contact person.
- 3) Click on participate button.
- 4) Click on Create Response Button.
- 5) Enter the Price in the Item tab.
- 6) Click on Technical RFx Button to upload the documents.
- 7) Finally click on Submit button before the Submission deadline date and time.

#### **PAY THE TENDER FEE & EARNEST MONEY DEPOSIT (EMD) AMOUNT**

Tender Fee & Earnest Money Deposit may be submitted to TVNL contact person in person or communicate to the contact person by email by attaching the scanned copy of the Demand draft/BG so that the contact person will allow access to open the RFx details and initiate bidding at TVNL website and attach the details of Part-I & fill the details of Part- II.

Inform TVNL contact person to provide the access to RFx to participate in the Bidding.

#### **OPEN THE RFx NUMBER IN THE VENDOR PORTAL**

**Vendor login:** Login to the TVNL Vendor Portal through the URL mentioned below.

Vendor Login URL: <http://webdispatcher.tvnl.in/srmprd>

Login Page shall appear on Vendor Portal page.

Step-1: Enter the User ID and Password and Change the Password during the first login.

Step-2: Click on Logon button

Step-3: Click on **RFx and Auctions**.

Click on Event Number to open the required RFx No. For E.g. 1000000XXX

If Bidder is accessing the bidder portal for the first time then system will show the Register button once you have opened the RFx as shown in the below screen.

***For the second time onwards system will not ask for bidder to Register.***

After click on Register button, if bidder is not paid the “**Tender fee amount**” to TVNL then system will show the error message as shown in the below screen.

**If you get the error message follow the below step:**

- 1) First vendor will pay “**Tender fee & Earnest Money Deposit**” amount and inform the same to TVNL contractor to provide the access to RFx.

**Note: Once contact person provide the access to RFx, then bidder will refresh button of the RFx and proceed further.**

After the bidder Register, system will show the below options to **Participate, Do Not Participate, Tentative and Create Response**.

**What is “Participate” button?**

- If bidder click on participate that means, Bidder interested in participating the bidding.

**What is “Do Not Participate” ?**

- If bidder click on Do Not Participate that means, Bidder is not interested to participate in the bidding.

**What is “Tentative” button?**

- If bidder click on Tentative that means, Bidder either will participate or not participate in the bidding.

**What is “Create Response” button?**

- After click on participate/Tentative button, bidder will use the create response to provide his/her bid quotation.

**TO CREATE THE RESPONSE FOR THE PARTICULAR RFx**

Step-1: Open the RFx.

Step-2: Click on Create Response.

**ANSWER THE QUESTIONS IN THE RFx**

After clicking on create Response button, system will show the error message. At this time, bidder will follow below steps:-

- 1) Click on Questions tab in the screen
- 2) Bidder will find Questions
- 3) Answer to those questions which are marked “\*” is mandatory.
- 4) Finally click on check button to check for any error message.

**ENTER THE PRICE OF THE ITEMS IN THE RFx RESPONSE**

Enter the Price for all the line items and click on Check if in case of errors and click on Submit to complete the RFx Response.

**ASK QUESTIONS TO TVNL IF YOU HAVE ANY DOUBTS REGARDING THE RFx**

As a Bidder if You have some questions related to RFx and bidder can also ask questions before submitting the bid.

**Step-1:** Click on “Questions and Answers (0)” button.

**Step-2:** Enter the Question and click on Send button to send the question to purchaser/TVNL. Vendor enters the Message and clicks on “Send” button.

**TO UPLOAD THE DOCUMENTS IN THE RFX RESPONSE AS A BIDDER**

Step-1: Click on “Notes and Attachments” tab

Step-2: Click on “Collaboration Room” as shown in the below screen to upload the documents.

Step-3: Click on Bidder Document folder.

Step-4: Click on Create Button.

Step-5: Select the Documents to upload the documents

Step-6: Enter the Name, Description of the document and click on Continue

Step-7: Click on Browser button to select the document in your local system.

Step-8: Once the document is selected click on save button.

Step-9: The saved document will be shown.

Step-10: Click on Logout and close the window once you have uploaded all the documents.

**TO SUBMIT THE RFx RESPONSE AS A BIDDER**

Once you have completed the Price, uploaded the documents click on check button for any errors.

If there is no errors system will show the below message.

Click on Submit button to complete the bidding.

After submitted the system will show the RFx Response status as “Submitted”.

**You can change the RFx Response even after bid has been submitted only when the submission date or time is still remaining (Not passed).**

Step-1: Open the RFx Response

Step-2: Click on Edit Button

Step-3: Make changes in the RFx Response and Click on Submit.

**You can also click in the Withdraw button to withdraw a bid and then If you Click on Edit button after withdrawal you can again make changes and resubmit the RFx Response if the submission time is remaining.**

### **ONLINE PAYMENT PROCESS**

After registering against the RFx you have to follow the below steps:

1. Go to TVNL website <http://tvnl.in>
2. Click on the navigation tab “ Tender Payment” link (at the Top Right hand side of the TVNL web home page).
3. It will redirect you to the online payment page.
4. Enter your User Id to proceed further.
5. Enter the RFx no. for which you want to participate.
6. Click on the “Go” button.
7. Click on the Pay button to proceed for payment of “Tender Fee” or “EMD”.
8. It will redirect you to the payment gateway for payment.
9. Enter your card details or go for online banking for payment.

(Signature & Seal of Bidder)



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JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S.Dhurwa, Ranchi-04, (Jharkhand)

## GENERAL TERMS & CONDITIONS

1. **Validity of Offers:** The bid submitted by the bidders should remain valid for a minimum period of 120 days from the date of opening of Part-I unless otherwise specified.
2. **Prices:** - The Bidder shall quote their rate keeping in view that the rate will remain firm for the entire contract period of 02(two) years. No extra amount against increase in works/any other by Govt. will be payable by TVNL to the contractor during the entire contract period.
3. **Late Receipt of the Tender:** Special attention of the bidder is invited about the receipt of the tender in time. Tenders received after the stipulated date and time due to any reason, whatsoever, shall not be considered. Bidders are requested to see that their bids are delivered in the office of TVNL before the prescribed date and time.
4. **Inspection of Site:** Before quoting the rate, bidders are advised to visit the work site and get satisfied with scope of work for which rates are to be quoted. Engineer -in -charge may be consulted for any clarification regarding scope of work and terms of tender. Bidder should go through the terms and conditions and technical specifications of work carefully and sign in all pages of the tender document.
5. **Instruction to Bidder:**
  - a. Bidders are advised to submit their bids strictly based on the scope of work, terms and conditions contained in the bid document and subsequent revisions/amendments, if any. The bid should be submitted along with covering letter.
  - b. The bid shall be prepared and submitted by typing or printing in English on white paper in consecutively numbered pages duly signed by the authorized signatory with company seal affixed on each page. Any part of the bid, which is not specifically signed by the authorized signatory and not affixed with company seal, may not be considered for the purpose of evaluation.
  - c. Copy of bid document, amendments/revisions to bid document if any, shall be signed and submitted along with the bid.
  - d. Complete bid document including all enclosures should preferably be submitted in hard bond or spiral binding and all pages should be numbered and must be signed by the authorized signatory with seal of the company.
  - e. Incomplete tender or tenders not submitted as per requirement as indicated in the NIT are likely to be rejected.
  - f. Bidders shall submit their offer strictly as per terms and conditions of the tender document without any deviation. If at any time any of the documents/information submitted by the bidder is found to be incorrect, false or untruthful, the bid and/or the resultant order may be summarily rejected/ cancelled at the risk of the bidder.
  - g. Failure to furnish all information and documentary evidence as stipulated in the bid document or submission of an offer that is not substantially responsive to the bid document in all respects shall be summarily rejected.
  - h. All bids will be received in duly sealed cover within the due date and time. Bids received after the due date and time is liable for outright rejection.
  - i. Variation may occur in the quantity of items required to be executed under this contract depending on the condition of machine and availability of units.
6. **Earnest Money Deposit (EMD):** Offers shall be accompanied with Earnest Money Deposit (EMD) of requisite value as specified in a separate envelope failing which offers are liable for rejection. EMD may be furnished in the form of Bank Guarantee from any Indian Nationalized Bank/Schedule Commercial Bank in favour of "TENUGHAT VIDYUT NIGAM LIMITED" payable at Ranchi. The Bank Guarantee should remain valid for 12 months or may be

deposited through payment gateway link of TVNL website. The earnest money will be forfeited on revocation of tender before expiry of validity of the tender or on refusal to enter into the contract agreement after the award is made by TVNL to the bidder.

Tenders without Earnest Money shall not be considered other than NSIC/SSI (Jharkhand) registered units. A copy of such registration has to be furnished for claiming exemption from EMD. Exemption from the Earnest Money to NSIC/SSI units will be considered as per Govt Circulation. SSI units of Jharkhand will be given preference as per Govt rule.

- 7. Security Deposit:** The successful bidder shall have to deposit total security deposit @ **10%** of order value in which initial security money @**5%** of the order value (including earnest money) will be retained as security deposit. Initial security money will be accepted as Bank Draft/ Bank Guarantee in the prescribed format not adjustable against running bill. The security deposit shall have to deposit well before execution of the Agreement. Balance **5%** of security money will be deducted from R/A bills. The security money will be released after successful completion of work and closure of contract.
- 8. Payment Terms:** Payment of Contractor on actual measurement basis will be made through running bills which will be verified by Engineer -in-Charge after successful completion of the work and payment of wages to the employee by the Contractor. The bill in triplicate will be submitted by the Contractor to the Engineer -In-Charge along with Bank Statement and Payment sheet showing payment of wages to workmen and other relevant documents like EPF & GST deposit slip etc before release of payment.
- 9. Paying Authority:** The Accounts Officer, TTPS, Lalpania will be the Paying Authority.
- 10. Engineer-in-Charge:** Concerned Electrical Executive Engineer will be the Engineer-in-Charge for this work. The works are to be executed as per direction of Engineer-in-Charge. Anything not mentioned here in shall be decided by the Engineer- In- charge of the work and shall be binding on the contractor.
- 11. Period of Contract:** The period of contract shall be 02 (two) years from the date as shall be mentioned in the work order.
- 12. Safety Measures:** The Contractor shall observe all the safety measures. Safety appliances have to be arranged by the Contractor at his own cost. Any compensation payable to the Contractor's employee in the event of accident or causality has to borne by the Contractor himself and TVNL will not be responsible on this accounts in any way. In case of any damage due to negligence mishandling and mal operation by the staff of contractor, the damage would be made good by the contractor at his own cost. In case of non compliance, proportionate deduction will be made from the bills of the contractor. The contractor shall provide all safety appliances, tools and equipments to his staff and ensure that those items will use by all staff while working and on whole duty period. Name of such important items are given below:-
  - i. Asbestos hand gloves, Apron Asbestos cloths etc.
  - ii. Rubber & Leather hand gloves,
  - iii. Safety shoes for all types of works,
  - iv. Safety belts and hooks,
  - v. Helmets, Ear muffs, Safety goggles, Nose muffs etc.
  - vi. Any other items related to safety rules may be assumed in contractor's scop
- 13. Scope of the work:** Scope of the work is attached here as a guideline to execute the work. However contractor has to execute all the work as per direction of Engineer-in-Charge for proper working of equipments.
- 14. Labour Laws:** The bidder has to follow the guidelines laid under the labour law, workmen compensation, and EPF rules in respect of minimum wages, EPF payment and other service conditions. In the event of any accident / casualty to the employees of contractor, the compensation payable to the employees is to be borne by the contractor and TVNL will not be responsible in any way.

- 15. Minimum Wage Guarantee:** The contractor shall strictly ensure that the minimum wages as stipulated by GoJ from time to time is paid to their workmen. Any increase in minimum wages during the period of contract, will have to be paid to their workmen without any claim on TTPS/TVNL. If the contractor fails to comply with guarantee of minimum wages then the agreement is liable to be terminated and security money may be forfeited.
- 16. Taxes:** All statutory taxes as applicable shall be deducted from each running bill of contractor.
- 17. Violation of Terms & Conditions of Tender:** All terms and conditions of tender shall form a part of the contract. If contractor fails to execute the work up to entire satisfaction of Engineer-in -Charge as per terms & conditions of tender, then the contract agreement can be terminated on one month prior notice and security money shall be forfeited.
- 18. Subletting:** The successful bidder will have to execute the work directly. No subletting will be allowed.
- 19. Competent Authority:** During the execution of work in case of any dispute between the Engineer-in -Charge and contractor, the decision of General Manager cum Chief Engineer Tenughat TPS will be final and binding.
- 20. Man Power:** A list of manpower arranged by the contractor should be submitted to Engineer-in -Charge of the work before deployment and same should be deployed with due permission of Engineer-in -Charge. If the performance of the deployed person is not found up to the mark at any time, the contractor will have to replace them with experience and suitable person. During the period of absence of contractors personnel TVNL will have right to deduct penalty from contractor's bill.
- 21. Tools & Tackles:** The contractor should have to arrange all tools & tackles as per scope of work whereas heavy tools like hydra crane will be arranged by TTPS if available as per request of contractor. Contractor will arrange standard make tools after getting approval from EIC. Contractor will arrange welding M/C, gas cutting set etc to carry out any mechanical work in maintenance of field instruments as and when required. The entry of tools and tackles inside the power house should be verified by the CISF and should match with the requirement. Pro-rata deduction will be made if any mismatch is found. Agency must arrange 02 nos. of master levels, 02 nos. of dial gauge and 02 nos. of vernier, sufficient nos. of filler gauge for leveling and measurement job at site. All above mentioned equipments must be in good condition.
- 22. Consumables:** The contractor has to provide all consumables like gas, electrodes, cotton waste, markin cloth, rustolin, K-Oil, Petrol/Diesel and abrasive materials at their own cost, if required and applicable. The entry of consumables inside the power house should be verified by the CISF and should match with the requirement. Pro-rata deduction will be made if any mismatch is found.
- 23. Site Cleaning:** The Contractor must keep the working areaclean after completion of the job. Waste materials and garbage shall be disposed off by the contractor at suitable place as per the instruction of the Engineer-in-Charge. Old unserviceable spare parts if so, has to be returned by the contractor to the Central Store/Site Store of TTPS as per the instruction of the Engineer-in-Charge.
- 24. Facilities to be provided by TVNL:**
- Unfurnished accommodation for office with water and electricity inside the power house will be provided free of cost if available.
  - The contractor shall be allowed to avail the facilities to extent available at our Chemical Lab at TTPS.
  - Guest house facility shall be provided to the visiting engineers/officers of the Contractor on chargeable basis if available.
  - Unfurnished accommodation may be provided to the employees of the Contractor, if available during the period of execution of work on rental basis as per departmental rule.
  - Electricity will be provided on chargeable basis as per JBVNL Tariff.

- f. Medical facility as available at project site shall also be extended to the employee of the contractor. No payment will be made for any medicine purchased from outside or any treatment or test done outside the TTPS Hospital.

- 25. Working Administration:** In general working hours of the TTPS employee will be the working hours of employee of Contractor. However in case of emergency or as per requirement of the job, contractor's employees will be required to work beyond working hours and also on rest days and holidays on the direction of Engineer -in -Charge. Employees deployed by the contractor will be under the operational control of Engineer -in -charge. In case of any employee working in shift/general duty falling sick or proceeding on leave the facts shall be brought to the knowledge of Engineer -in -Charge and deficit will be made by the contractor either by the existing staff by performing overtime duty or getting other manpower without adding any additional cost to the Project. If the shift remains vacant the penalty will be deducted from the contractor's bill. It shall be binding on contractor that it does not directly or through its personnel divulge the secrecy of the establishment of TTPS. Any person deployed by contractor will not be a member of any trade union operating in the TTPS Projects establishment or outside. They will not take part in any strike or sympathized with any such movement.
- 26. Penalty:** In case of any breakdown, fault or emergency maintenance, the contractor has to attend the same without any delay. In case of failing or deliberated neglect, the work will be carried out through outside agency at the cost & risk of the contractor and this may be liable for termination of their contract. A poor (Manpower: skilled or unskilled, T&P and consumables) mobilization during OH or Emergency jobs shall attract a penalty of Rs 5000/ per day. Further any safety lapses like non use of PPEs, improper handling of gas cylinders, improper power connection, damaged electrical equipments, non use of plug tops, inadequate scaffolding where work is being carried out at height, poor/absence of temporary fencing where railing has been cut to facilitate lifting up or/and down of materials etc shall attract a penalty of Rs 5000/ per occasion. Scrap generated during the work shall be removed every day(as directed by EIC) by the contractor to keep the area clean. If scrap is not removed then the same will be removed at risk and cost of agency @ Rs 5000/MT will be deducted as penalty.
- 27. Departmental Facilities:** In case of major repair of equipments and testing of special equipments contractor should have to provide services of an expert from manufacturer and special tools & tackles. The contractor will also provide all other technical staffs and T&P's consumables etc free of cost. However, all spares parts shall be provided by TVNL on free of cost.
- 28. Transportation:** Successful bidder will be required to arrange his own conveyance for their employee for attending their duties.
- 29. Mobilization Time:** The successful bidder will be given maximum 15 days time from the date of LOI / Work Order for Mobilization of resources, manpower to start the work. However for the work under ARC, contractor shall be given the mobilization period of one day. The contractor should be in position to ensure mobilization even within shorter period to start the work if need so arises like availability of the shutdown of units. No extra claim shall be entertained by either for idle labour or reduced mobilization period.
- 30 Agreement:** The firm or his legally authorized representative having Power of Attorney will have to execute an agreement with the ordering authority within 25 days of issue of order in the event of placement of order on them. The agreement will have to be prepared on a non-judicial stamp paper of Rs.100/- only on Nigam's format. Bank Guarantee against Security Deposit, Performance Guarantee etc. as per order will have to be furnished well before execution of agreement. The contractor is required to produce all original paper and documents for verification before agreement failing which order will be cancelled. Contractor will not be allowed to start the work without executing the agreement.
- 31 Jurisdiction of Court:** Any dispute between the Contractor and TVNL shall be settled in the Civil Court, Ranchi, Jharkhand only.

(Signature & Seal of Bidder)



## TENUGHAT VIDYUT NIGAM LIMITED

### तेनुघाट विद्युत निगम लिमिटेड

(A Govt. of Jharkhand Undertaking)

JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S.Dhurwa, Ranchi-04, (Jharkhand)

**NIT No. 27/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1000014214**

**Sub:** Annual Rate Contract for 02 (Two) years and annual overhauling of Boiler rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) of Unit-1&2 of TTPS, Lalpania.

#### **SCOPE OF WORK**

All works must be completed within schedule time frame allotted by Engineer/Officer in charge. Depute an eligible supervisor to look & carry out the work efficiently.

#### 1. Servicing/Replacement of Brg. of seal air fan.

- Decoupling of the seal air fan from motor.
- Removal of top half of bearing housings.
- Checking bearing clearances.
- Removal of lower part of bearing housing.
- Taking out the damaged bearings by suitable puller.
- Fitting new bearing with sleeves.
- Apply grease on bearing.
- Placing the bearing housing.
- Opening of inspection doors and cleaning of Impeller.
- Alignment of the fan with motor and coupling of motor with fan.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 2. OH/Replacement of seal air fan impeller assembly.

- Take PTW from the area engineer for attending the defect, dismantling of duct pipe.
- Opening of inspection doors and checking for freeness of impeller.
- Decouple motor & fan and shift Motor to facilitate shaft removal.
- Dismantling of bearing housing of seal air fan impeller
- Dismantle impeller hub& Remove bearings.
- Replace/rectify shaft/impeller Box-up impeller, Labyrinth seals, bearings etc.
- Proper alignment to be done before coupling for trial.
- Trial run of seal air fan and measurement of vibration and Temperature.
- Taking corrective actions for rectification of any abnormality in vibration and temperature.
- Removal Of waste and scrap materials immediately from site.
- Checking and repairing of Flappers

#### 3. Replacement of impeller wears plate of PA Fan

- Transportation of new wear plate from stores to site.
- Opening of manholes.

- Removal of old wears plates.
- Replacement of wear plates with new ones (new plates & bolts).
- Closing of manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 4. Replacement of damaged spiral casing liners of PA Fan

- Transportation of new liners from stores to site.
- Opening of manhole.
- Removal of old liners by cutting bolts.
- Fitting of new liners and bolting it with casing and tack welding of nuts.
- Removal of waste and scrap materials from inside of fan.
- Closing of all Manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 5. Replacement of DE bearing of PA Fan

- Decoupling of fan from motor.
- Dismantling of lube oil outlet connection from bearing and their thorough cleaning.
- Removal of PA Fan motor from its Foundation.
- Taking out the coupling half from the fan side.
- Removal of top cover of bearing housing of fan DE side.
- Checking bearing clearances.
- Removal of bearings by a suitable puller mechanism.
- Removal of bearing housing lower half of DE
- Polishing of bearing portion of shaft by very fine emery paper.
- Fitting new bearings after fitting the lower portion of bearing housing.
- Box up of the bearings.
- Connecting oil inlet and outlet pipes.
- Fitting of fan side coupling half.
- Placing the motor on foundation and coupling with fan after alignment.
- Coupling bushes & bolts, replaced if found damaged.
- Trial run of the fan.
- Checking for temp of bearings and any oil leakage and corrective measure till successful operation of the fan.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 6. Replacement of NDE bearing of PA fan

- Decoupling of fan from motor.
- Dismantling of lube oil inlet & outlet connection and their thorough cleaning.
- Removal of top cover of bearings of fan NDE side.
- Checking bearing clearances.
- Removal of bearing housing lower half of DE.
- Removal of bearings by a suitable puller mechanism.
- Polishing of bearing portion of shaft by very fine emery paper.
- Fitting new bearings after fitting the lower portion of bearing housing.

- Box up of the bearings.
- Connecting oil inlet and outlet pipes.
- Coupling with motor after alignment.
- Coupling bushes & bolts replaced if found damaged.
- Checking for temp of bearings and any oil leakage and corrective measure till successful operation of the fan.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 7. Replacement of coupling bushes & bolts of PA fan

- Removal of coupling bolts.
- Taking out the Coupling bush from fan side coupling.
- Checking Coupling bush and bolt and replace with newer one if found damaged.
- Fitting coupling bush and bolts and their tightening.
- Clean the serviced equipments after completion of work.

#### 8. Replacement of damaged coupling of PA fan

- Decoupling of fan from motor.
- Removal of PA Fan motor from its foundation.
- Taking out the coupling half from the fan side by heating.
- Cleaning of key slot of the shaft.
- Heating of the new coupling half, fitting it on the shaft.
- Placing the motor on foundation and coupling with fan after alignment.  
Coupling bushes & bolts replaced if found damaged.
- Trial run of the fan.
- Clean the serviced equipments after completion of work.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 9. Complete overhauling of L.O.P system of PAF.

- Removal of oil and water connections of the cooler.
- Loosening of the foundation bolts.
- Removal of the defective cooler.
- Placing new/repaired cooler.
- Connecting all the oil and water lines.
- Charging of the cooler from water side.
- Checking for any leakage any attending the same.
- Shifting of the defective cooler to a specified location.
- Drain oil from tank with suitable arrangement and store it.
- Dismantling of lube oil inlet and outlet connection from bearing housings and their thorough cleaning.
- Dismantling level indicator and oil return looking glass and cleaning of glass.
- Clean the oil tank.
- Pour storage oil into tank if seems clean otherwise Tank fill with new drum of oil.
- Connecting inlet and outlet pipe after cleaning through air.

- Checking coupling of LOPs and cleaning of suction and discharge filter.
- Trial run of lops
- Attended oil leakages if any.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 10. PA FAN-Replacement of impeller

- Decoupling of fan from motor.
- Dismantling of lube oil inlet& outlet connection and their thorough cleaning.
- Removal of top cover of bearings of fan DE & NDE side.
- Checking bearing clearances.
- Removal of labyrinth seal & impeller seal.
- Removal of bearing housing lower half of DE & NDE side.
- Dismantling & removal of top cover.
- Removal of rotor assembly.
- Fixing of New/Repaired Impeller, suction cone (after blue matching), wear plate, coupling half, seal ring, spacer rings, bearings & etc on Rotor.
- Fixing of lower half of bearing housing with base after blue matching.
- Erection of new assembly.
- Erection of top cover.
- Inspection of Tie rod, Labyrinth seal & Impeller seal plate and replaced if found damaged.
- Fitting of impeller seal plate & Labyrinth seal.
- Box up of the bearing housing.
- Connecting oil inlet and outlet pipes.
- Coupling with motor after alignment.
- Coupling bushes & bolts replaced if found damaged.
- Trial run of the fan.
- Checking for temp. of bearings and any oil leakage and corrective measure till successful operation of the fan.
- Transportation of taken out Rotor Assembly to place decided by EIC.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.
- Dismantling of Impeller, suction cone, coupling half, seal ring, spacer rings, bearings & etc from taken out Rotor as per instruction of EIC.

#### 11. PA FAN-Repairing of PA impeller

- Opening of manholes.
- Inspection of Impeller.
- Remove wear plate
- Cleaning of Impeller as per requirement.
- Filling/Remetalling of damaged Portion by Lo-Therm-468 welding electrode.
- Grinding of Impeller as per requirement.
- Fixing of wear plate.
- Closing of manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 12. PA FAN-Repairing of casing & adjoining duct

- Opening Manholes of PA Fan Casing & duct as per requirement.
- Making temporary arrangement of lighting & scaffolding identifying damaged portion, cutting plates of required size for patch work, welding of plates.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.
- Closing of Manholes.

#### 13. Internal Checking of PAF and its brgs.

- Opening all man holes.
- Inspection of impeller, impeller seal plate and wear plate.
- Cleaning of Impeller.
- Tightening wear plate bolts.
- Dismantling of lube oil outlet connection from bearing and their thorough cleaning.
- Removal of top cover of bearing housing of fan DE & NDE side and checking bearing clearance.
- Checking /tightening bolts of bearing housing.
- Inspection of suction cone and tightening its bolt if required.
- Closing all man holes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 14. Assistance of Manpower during Vibration analysis of Rotary Equipments.

- Engage Manpower during Vibration analysis of Rotary Equipments and perform work as per instruction of EIC.

#### 15. PAF- REPLACEMENT OF MOTHER CASING PLATE.

- Transportation of new mother casing plate of PA Fan from central store to site.
- Opening of all manholes, making temporary arrangements of light & scaffolding.
- Remove Spiral Casing Liners.
- Cutting & removing mother casing plate.
- Grinding new mother casing plate and casing of PA Fan for welding.
- Welding mother casing plate one by one.
- Fitting Spiral Casing Liners.
- Remove all scraps.
- Closing of manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 16. FD-FAN INTERNAL INSPECTION & CHECKING.

- Opening of fan suction, discharge and servomotor manhole doors.
- Silencer inspection.
- Opening of shaft protection tubes & bullet cover and checking for any

- Oil leakages in Lub oil servomotor & main bearing assembly.
- Assistance in checking & replacement of C&I points like thermocouple, RTD, vibration pick up.
- Blade cleanings & assisting in DPT of blades either in position or by removing the blades.
- Cleaning & greasing of servomotor operating levers and bearings.
- Blades pitch operation checking & adjustment if required.
- Removal of coupling guards, checking of motor to fan alignment & refitting of coupling guards.
- Bullet covers and shaft protection tubes placement and bolt tightening after providing packing ropes.
- Bolt tightness checking, measuring blade clearance with respect to casing and impeller hub and adjustment if required.
- Cleaning of fan suction & discharge casing and closing of manhole doors.
- Assistance in trial run and attending defect if any.
- Cleaning of the serviced equipments should be done after completion of the maintenance immediately.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 17. FD FAN- BEARING REPLACEMENT.

- Removal of distance piece, fan decoupling and removal of insulation of diffuser cone if required.
- Removal of C&I connection, servomotor connections and LUB oil
- Connections from the bearing housing by removing bullet cover to Upper half if not removed earlier.
- Removal of suction & delivery side expansion joints of diffuser.
- Pulling of diffuser inside discharge duct.
- Removal of impeller blades.
- Dismantling of impeller hub with servomotor assembly from main bearing assembly and lifting of the same from the position & shifting it to maintenance bay
- Removal of the main bearing assembly from position and shifting it to maintenance bay.
- Removal of coupling hub from the shaft.
- Removal of covers & dismantling of the simmer rings, inner rings, o rings & the bearings from the bearing housing & shaft assembly.
- Cleaning of dismantled components and assembled with new or old o ring, simmer ring, inner ring & bearing as per requirement.
- Box up of bearing housing by providing suitable gaskets and checking of axial floats of the assembly and adjustment if required.
- Cleaning of impeller hub assembly & servo motor assembly and oil top up and pressure testing in blade bearings.

- Assembling the impeller hub & servo motor assembly. Operation checking & shifting it to site.
- Lifting & positioning of the main bearing assembly to fan casing and leveling and bolt tightening.
- Assembling the impeller hub & servomotor assembly to main bearing assembly.
- Pulling back of diffuser cone & providing expansion joints.
- Restoration of servomotor connection, C&I connections & LUB oil connections.
- Fixing of blades & torque tightening bolts.
- Servomotor operational checking & adjustment if required.
- Coupling of fan pulling back shaft protection tube by providing rope & fixing of coupling guards.
- Cleaning of the serviced equipments should be done after completion of the maintenance immediately.
- Cleaning site and dispose waste & scrap materials to scrap yard.

#### 18. FD FAN: COUPLING & ALIGNMENT

- Decoupling the fan and fittings of clamps and dial gauges.
- Loosening of motor foundation bolts for providing shims.
- Taking readings and providing shims to adjust and align the motor to the specified value.
- Tightening the foundation bolts and taking the final alignment readings.
- Assisting in de-coupled trial run (if any).
- Coupling the motor with fan and putting back coupling guard.
- Cleaning of the serviced equipments should be done after completion of the maintenance immediately.
- Cleaning site and dispose waste & scrap materials to scrap yard.

#### 19. FD fan-Replacement of pitch control blades.

- Removal of distance piece, fan decoupling and removal of insulation of diffuser cone if required.
- Removal of C&I connection, servomotor connections and Lub oil connections from the bearing housing by removing bullet cover top half if not removed earlier.
- Removal of suction & delivery side expansion joints of diffuser.
- Pulling of diffuser inside discharge duct.
- Removal of pitch control blades.
- Fixing of blades & torque tightening bolts.
- Servomotor operational checking & adjustment if required
- Cleaning of the serviced equipments should be done after
- Completion of the maintenance immediately.

#### 20. ID FAN- INT.REP. & COUPLING CHECKING.

- Maintenance of ID fan including coupling and bearing checking and alignment.

- Opening of all the manholes
- Removal of accumulated dust/oils from inside & outside of the fans.
- Checking the condition of impeller, its hub casing, casing liners, suction cones.
- Revisioning of DE & NDE bearing
- Replacement of complete oil of bearing.
- Checking and replacement of oil catcher (if required).
- Revisioning of both couplings.
- Removal of complete old grease.
- Washing of coupling and applying fresh grease.
- Replacement of coupling bolts, checking water hose pipes and associated valves and their replacement.
- Attending any leakages from water as well as oil circuit, backwashing of coolers.
- Checking the alignment/run out of fans.
- Replacement of damaged hoses of DE/NDE bearings.
- Attending minor leakages from hydro coupling.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 21. Repair of casing and adjoining ducts of ID Fan.

- Opening of manholes.
- Identifying damaged portion.
- Removal of damaged portion by gas cutting.
- Repair of damaged portion by MS welding.
- Clean the work place (inside of casing).
- Closing of manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 22. Replacement of impeller wears plate of ID Fan.

- Transportation of new wear plate from central stores to site.
- Opening of manhole doors.
- Removal of old wears plates.
- Replacement of wear plates with new ones (new plates & bolts)
- Clean the work place (inside of casing).
- Closing of manhole door.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 23. Replacement of damaged spiral casing liners of ID Fan.

- Transportation of new liners from central stores to site.
- Opening of manhole doors
- Removal of old liners by cutting bolts.
- Fitting of new liners and bolting it with casing and tack welding of nuts.
- Clean the work place (inside of casing).

- Closing of manhole door.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 24. Servicing /OH of hydro coupling.

- Thorough cleaning of the hydro coupling.
- Draining the oil from the hydro coupling.
- Loosening the parting plane bolts.
- Opening the top cover.
- Dismantling the rotary parts and their cleaning.
- Taking measurements of all the bearings, replacement of damaged bearings, lub oil pump, pinion, reassembly of the parts, filling of oil.
- Alignment of ID Fan-HC-Motor.
- Taking of trial run.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 25. Checking and replacement of LOP of hydro coupling of ID Fan

- Opening of the inspection door of the hydro coupling
- Dismantling of the damaged pump with drive pinion.
- Revisioning of the spur gear mounted on the hydro coupling wheel, fitting of a repaired lube oil pump and pinion.
- Closing of the hydro coupling with new gasket and successful trial of the ID fan.
- Clean the work place (Inside of casing).
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 26. Replacement of damaged DE/NDE side bearing in ID Fan

- Isolate water line connections of bearings.
- Remove of top cover of the bearing.
- Loosening the bearing tightening bolts.
- Removal of the top half of the bearing.
- Jacking of the rotor on both DE and NDE side by applying hydraulic jack.
- Removal of lower portion of the bearing carefully so as not to damage the shaft portion.
- Removal of complete oil from the bearing housing.
- Smoothing of bearing portion of the shaft.
- Shifting of new bearing to site.
- Fitting of lower portion of the bearing.
- Removal of jack.
- Fitting of both the oil catcher rings, fitting of upper half of the bearing, tightening both half of the bearing.
- Connect the cooling water line.
- Filling up oil in the bearing.
- Successful trial of the fan.

- Clean inside of casing.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 27. Replacement of defective coupling of ID Fan

- Removal of all the coupling bolts.
- Taking out the distance piece.
- Shifting of motor.
- Heating the coupling hub by gas heating.
- Taking out the defective coupling.
- Taking measurement of shaft dia.
- Taking measurement of new coupling bore and finishing of internal surface of the coupling.
- Fitting of new coupling,
- Alignment of the hydro coupling and motor/fan.
- Coupling of the hydro coupling and motor/fan.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

#### 28. ID FAN: ALIGNMENT.

- Decoupling the fan and fitting of clamp and dial gauges.
- Loosening of motor foundation bolts for providing shims.
- Shifting of motor and modification of motor base if any required
- Taking readings and providing shims in pedestal to adjust and align the motor to the specified value.
- Tightening of foundation bolts and taking final alignment readings.
- Final tightening of bolts and final checking of air gap.
- Assisting in de-coupled trial run (if any)
- Putting back the end cover.
- Cleaning of the serviced equipments should be done after completion of the maintenance immediately.

#### 29. ID Fan-Replacement of impeller.

- Decoupling of fan from HC and motor from HC (If required).
- Dismantling of water inlet & outlet connection from both side DE & NDE.
- Removal of top half of DE & NDE side bearings & bearing housing of fan.
- Removal of labyrinth seal & impeller seal.
- Removal of complete oil from the bearing housings.
- Removal of lower half of DE & NDE side bearing housing & bearing if required.
- Dismantling & removal of top cover.
- Removal of rotor assembly.
- Fixing of New/Repaired Impeller, suction cone (after blue matching), wear plate, Coupling & etc on Rotor.

- Fixing of lower half of bearing housings with base after blue matching and bearings if required.
- Erection of new assembly.
- Erection of top cover.
- Inspection of Labyrinth seal & Impeller seal plate and replaced if found damaged.
- Fitting of impeller seal plate & Labyrinth seal.
- Fitting of oil catcher rings, Thrust collar & etc, fitting of upper half of the bearings & bearing housing, tightening both half of the bearing.
- Connecting water inlet & outlet connection of both side DE & NDE.
- Filling up oil in the bearing.
- Alignment of the hydro coupling & motor and hydro coupling & fan.
- Coupling of the hydro coupling & motor and hydro coupling & fan.
- Trial run of the fan.
- Checking for temp of bearings and any water/oil leakages from HC and hose pipes and corrective measure till successful operation of the fan.
- Transportation of taken out Rotor Assembly to place, decided by EIC.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.
- Dismantling of Impeller, suction cone, coupling & etc from taken out Rotor as per instruction of EIC.

### 30. ID Fan-Repairing of impeller

- Opening of manholes.
- Inspection of Impeller.
- Remove wear plate
- Cleaning of Impeller as per requirement.
- Filling/Remetalling of damaged Portion by Lo-Therm-468 welding electrode.
- Grinding of Impeller as per requirement.
- Fixing of wear plate.
- Closing of manholes.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

### 31. OH & Replacement of ID Fan Hydro coupling.

- Thorough cleaning of the hydro coupling.
- Loosening the parting plane bolts.
- Opening the top cover.
- Perform work as per instruction of service Engineer engaged for Supervision.
- Shifting of overhauled HC to ID Fan area.
- Decouple existing HC from fan and motor side.
- Removing both side coupling, loosening all fixing bolts and cutting stoppers of existing HC and Taken out.
- Placing the overhauled HC, revisioning of both side coupling, fitting of both side coupling.

- Alignment of HC with motor and fan.
- Tightening of all foundation bolts, Coupling HC with motor and fan.
- Fixing of Power Cylinder, if Required.
- Topping up of Torque 10.
- Trial run of Fan.
- Placing the taken out HC at place decided by EIC.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

### 32. REPAIRING ACW LINE OF FD/ID/PA FAN AREA.

- Making temporary arrangement of lighting & scaffolding if required.
- Identifying damaged portion.
- Arresting water leakage temporary by using clamps.
- Cutting damaged pipes if required and replaced with newer ones of required size by welding.
- Replaced damaged vent valves with newer one.
- Cleaning site and Dispose waste & Scrap materials to Scrap yard.

### 33. Replacement of labyrinth Seal

- Remove the existing Labyrinth Seal by un-bolting, lifting the shaft by Jack as required.
- Remove all Gaskets and placed new one of required shape and size, fixing & bolting.
- Install the labyrinth seal, bolting, tightening.
- Ensure the Fitting.



## TENUGHAT VIDYUT NIGAM LIMITED

### तेनुघाट विद्युत निगम लिमिटेड

(A Govt. of Jharkhand Undertaking)

JUPMI Building premises in ABD Area, Ranchi SmartCity, P.O&P.S.Dhurwa, Ranchi-04, (Jharkhand)

#### BILL OF QUANTITY

#### PART – II (PRICE PART)

**NIT No. 27/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1000014214**

**Sub:** - Annual Rate Contract for 02 (Two) years and annual overhauling of Boiler rotary Equipments (FD Fans, ID Fans, PA Fans, Seal Air Fans) of Unit-1&2 of TTPS, Lalpania.

| Sl. No.   | Description of work                        | Quantity | Rate (Rs.) | Amount (Rs.) |
|-----------|--------------------------------------------|----------|------------|--------------|
| <b>10</b> | ARC for 2 yrs & OH of Boiler rotary        | 1.000    |            |              |
| 10.1      | SAF-Servicing/Replacement of Bearing.      | 24.000   |            |              |
| 10.2      | SAF-OH/Replace of Seal Air Fans complete   | 24.000   |            |              |
| 10.3      | PA Fan-Replacement-impeller wear plates    | 16.000   |            |              |
| 10.4      | PA Fan-Replacement of spiral casing liner  | 12.000   |            |              |
| 10.5      | PA Fan -Replacement of DE side bearing     | 8.000    |            |              |
| 10.6      | PA Fan-Replacement of NDE side bearing     | 8.000    |            |              |
| 10.7      | PA Fan-Replacement of coupling bush& bolts | 24.000   |            |              |
| 10.8      | PA Fan -Replacement of coupling            | 12.000   |            |              |
| 10.9      | PA Fan-Complete OH of LO system of PAFs    | 24.000   |            |              |
| <b>11</b> | PA Fan-Replacement of impeller             | 6.000    |            |              |
| 11.1      | PA FAN-Repairing of PA impeller            | 70.000   |            |              |
| 11.2      | PA FAN-Repair of casing & adjoining duct   | 300.000  |            |              |
| 11.3      | PA Fan-Internal checking of Fan & brgs.    | 40.000   |            |              |
| 11.4      | Assistance in vibration analysis           | 30.000   |            |              |
| 11.5      | PAF-Replacement of Mother Casing Plate     | 3.000    |            |              |
| 11.6      | FD FAN-Internal Inspection & Checking      | 16.000   |            |              |
| 11.7      | FD FAN-Brg. Replacement                    | 4.000    |            |              |
| 11.8      | FD FAN-Coupling and alignment              | 12.000   |            |              |
| 11.9      | FD Fan-Replacement of pitchcontrol blade   | 4.000    |            |              |

|           |                                          |         |  |  |
|-----------|------------------------------------------|---------|--|--|
| <b>12</b> | ID Fan-Int. Rep. and coupling checking   | 24.000  |  |  |
| 12.1      | ID Fan-Repair of casing & adjoining duct | 100.000 |  |  |
| 12.2      | ID Fan Replacement of impeller wearplate | 6.000   |  |  |
| 12.3      | ID Fan Replacement casing liners         | 4.000   |  |  |
| 12.4      | ID Fan-servicing/OH of hydrocoupling     | 4.000   |  |  |
| 12.5      | ID Fan Replacement of pump of hydrocoupl | 4.000   |  |  |
| 12.6      | ID Fan Replacement of DE/NDE side brg    | 8.000   |  |  |
| 12.7      | ID Fan Replacement of coupling           | 8.000   |  |  |
| 12.8      | ID FAN-Alignment                         | 8.000   |  |  |
| 12.9      | ID Fan Replacement of impeller           | 2.000   |  |  |
| <b>13</b> | ID Fan-Replacement of ID impeller        | 70.000  |  |  |
| 13.1      | OH& Replacement of ID Fan HC             | 2.000   |  |  |
| 13.2      | Repairing of ACW Line of FD/ID/PAF Area. | 80.000  |  |  |
| 13.3      | Replacement of labyrinth Seal of ID/PA f | 20.000  |  |  |

**Part-II: - Price bid to be submitted in TVNL SRM website only.**

**(Certificate in Bank Letter Head)**

SOLVENCY CERTIFICATE

BANK NAME /BANK SEAL

BRANCH.....

DATE.....

To,

.....  
.....  
.....

This is to certify that to the best of our knowledge and information that  
Mr/Mrs/Ms/Company Name/Firm Name.....a customer of our bank  
having Account type.....and Number.....is respectable  
and can be treated solvent up to a sum of  
Rs.....(Rupees.....Only).

Branch Manager

Sign

Seal of Bank

**DRAFT**

**BANK GUARANTEE FOR EARNEST MONEY DEPOSIT**  
**(TO BE STAMPED IN ACCORDANCE WITH STAMP ACT)**

(TO BE ISSUED BY ANY NATIONALISED/SCHEDULED BANK AUTHORIZED BY RBI TO ISSUE: A BANK GUARANTEE)

Ref:

Bank Guarantee No:

Date:

To,

Tenughat Vidyut Nigam Limited

JUPMI Building Premises in ABD AREA, Smart City,

Dhurwa, Ranchi-834004 Jharkhand

Dear Sir,

In consideration of M/s Tenughat Vidyut Nigam Limited having its Registered Office at JUPMI Building Premises in ABD AREA, Smart City, Dhurwa, Ranchi- 834004 Jharkhand hereinafter called "the Nigam" which expression shall unless repugnant to the subject context or meaning thereof includes its successors administrators and assigns (having agreed under the terms and conditions contained in letter no. \_\_\_\_\_ issued in favour of M/s \_\_\_\_\_ for (hereinafter referred to as "the contract" to accept the deed of guarantee as herein provided for Rs. \_\_\_\_\_ from the Nationalized Bank as Earnest Money Deposit: to be made M/s \_\_\_\_\_ (hereinafter called "the Contractor, we the \_\_\_\_\_ Bank (hereinafter referred to as the said Bank) having its Registered Office at \_\_\_\_\_ do hereby undertake and agreed to pay the Nigam to the extent of Rs. \_\_\_\_\_ on demand stating that the amount claimed by the Nigam is due and payable by the contractor for the reasons.

In accordance with your Tender Notice for \_\_\_\_\_ undersigned specification no....., dated....., M/s. \_\_\_\_\_ (thereafter called the tenderer) we, the ..... (the Bank) submit said Bank Guarantee on the following terms & condition: -

- (i) Whereas to participate in the said tender for the following:
  - a. ....
  - b. ....
  - c. ....

It is condition in the tender documents that the tenderer has to deposit Earnest Money amounting to Rs..... (Rupees ..... ) with respect to the tender with M/s TENUGHAT VIDYUT NIGAM LIMITED (hereinafter referred to as 'Corporation') by a Bank Guarantee from a Nationalized/Scheduled Bank irrevocable and operative till 06 (six) months after the validity of the offer (i.e..... month from the

date of opening of tender) for the like amount which is likely to be forfeited on the happening of contingencies mentioned in the tender documents.

- (ii) We \_\_\_\_\_ Bank agree that the Nigam shall be the sole judge as to whether the said contractor has failed/ neglected in performing any of the terms and conditions of the said contract and the decision of the company in this behalf shall be final and binding on us.
- (iii) We the said Bank further agree that the Guarantee herewith contained shall remain in full force and effect up to \_\_\_\_\_ and any claim received after the said date shall in no case bind on the Bank.
- (iv) The right of the Corporation to recover the said amount of Rs..... (Rupees.....) from us in manner aforesaid will not be precluded/ affected even, if disputes have been raised by the said M/s ..... (Tenderer) and /or dispute or disputes are pending before any authority, office, tribunal and /or arbitrator (s) etc.
- (v) We \_\_\_\_\_ Bank lastly undertake not to revoke this guarantee during its currency except with the previous consent of the Nigam, in writing.
- (vi) The Bank has under its constitution, power to give this guarantee and Mr. Manager, who has signed it on behalf of the Bank, has authority to do so.
- (vii) This Bank Guarantee will not be discharged due to the change in the constitution of the Bank or the Bidder / Contractor.

Dated, this \_\_\_\_\_ day of \_\_\_\_\_ 200 \_\_\_\_\_

Signature of the authorized person

For and on behalf of the Bank

Name:

(With Bank's Rubber Stamp)

Attorney as per power of Attorney No:

Dated:

Witness:

Signature:

Name:

Official address:

Under Jurisdiction of Ranchi Court only.